

DAY 05: S3 STORAGE, IAM, LEAST PRIVILEGE PRINCIPLE & AWS POLICIES

Cloud Computing & DevOps Master Curriculum

1. Amazon S3 (Simple Storage Service)

Amazon S3 is an object storage service offering industry-leading scalability, data availability, security, and performance. Data is stored in **Buckets** as **Objects** (files up to 5TB).

- **S3 Standard:** High durability (99.999999999% - 11 9s), frequent access.
- **S3 Intelligent-Tiering:** Automatic cost savings by moving objects between access tiers based on usage patterns.
- **S3 Standard-IA & One Zone-IA:** Infrequent access with rapid retrieval.
- **S3 Glacier Flexible & Glacier Deep Archive:** Low-cost archive storage (retrieval times from minutes to hours).

2. AWS IAM (Identity & Access Management)

IAM securely controls access to AWS resources. Core concepts include:

- **IAM User:** An entity representing a person or service (has password / access keys).
- **IAM Group:** Collection of IAM users (assign policies to group for easy administration).
- **IAM Role:** An identity with permission policies that can be assumed by AWS services (e.g., EC2 assuming a role to read S3).
- **IAM Policy:** JSON document explicitly defining allowed/denied actions on specific resources.

3. Principle of Least Privilege & Policy Types

Principle of Least Privilege: Grant users and services ONLY the minimum permissions necessary to perform their required tasks.

Policy Type	Characteristics
AWS Managed Policies	Created and maintained by AWS (e.g., AdministratorAccess, ReadOnlyAccess).
Customer Managed Policies	Custom JSON policies created by you in your AWS account tailored to exact requirements.
Inline Policies	Policies embedded directly into a single IAM user, group, or role.

Example S3 Read-Only JSON Policy snippet:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": ["s3:GetObject", "s3:ListBucket"],
    }
  ]
}
```

```
    "Resource": ["arn:aws:s3:::my-app-bucket", "arn:aws:s3:::my-app-bucket/*"]
  }
]
```

KEY TAKEAWAYS & SUMMARY

- ✓ S3 offers 11 9s of durability and stores data object-by-object inside globally unique bucket names.
- ✓ IAM Roles eliminate hardcoded AWS secret keys on EC2 instances.
- ✓ Always apply the Least Privilege Principle to prevent unauthorized actions and security breaches.